

Night Light PCB – System Design & Architecture

1. Project Overview

This system is a **mains-powered smart night light** with:

- Integrated motion detection (PIR) and ambient light sensing
- Rechargeable battery backup (~24 hours operation)
- Wireless RF remote for override control

The system consists of two independent PCBs:

MAIN PCB (Night Light Unit)

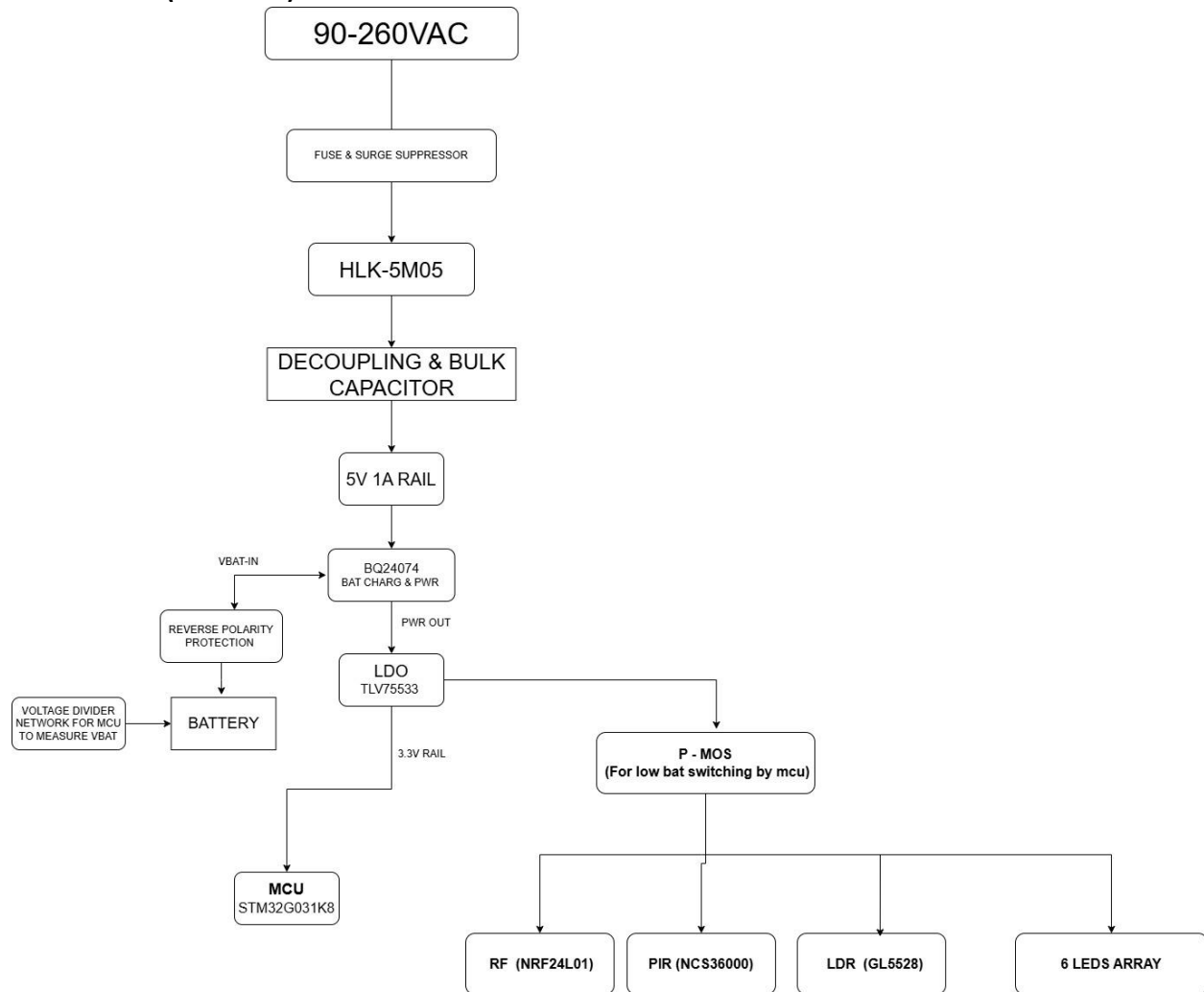
- Handles power conversion (240V AC → DC)
- Battery charging and backup
- Sensor processing
- LED driving
- RF reception

REMOTE PCB

- Battery powered
- Sends RF commands via button inputs

2. Component Architecture

Power Chain (Main PCB)



3. Component-Level Design

3.1 AC/DC Conversion — HLK-5M05

Role:

Converts 240V AC → isolated 5V DC

Why selected:

- Provides electrical isolation (critical for safety)
- Compact vs discrete flyback design
- Sufficient power (5W) for system + charging
- Reduces design complexity and certification risk

3.2 Charger + Power Path — BQ24074

Role:

- Charges Li-ion battery
- Powers system load when mains is present
- Seamlessly switches to battery during outage

Why selected:

- Integrated power-path management (no external OR-ing required)
- Input current limiting → prevents overload of AC/DC module
- Thermal regulation → safe in compact enclosure
- Stable system rail during transitions (important for MCU + RF)

3.3 Battery — 18650 Li-ion (3000mAh)

Role:

Backup energy storage

Why selected:

- Meets 24-hour runtime requirement
- High energy density for compact design
- Compatible with simple charging architecture

3.4 3.3V Regulation — TLV75533

Role:

Provides stable 3.3V rail for logic and RF

Why selected:

- Superior transient response (handles RF current spikes)
- Low output noise → critical for PIR and RF stability
- Low dropout suitable for Li-ion supply range
- Improves system reliability vs simpler LDOs

3.5 MCU — STM32G031K8

Role:

Central system controller

Responsibilities:

- State machine execution
- RF communication handling
- Sensor processing
- LED control

Why selected:

- Low power operation
- Integrated ADC (for LDR)
- SPI interface (for RF module)
- Sufficient processing headroom

3.6 RF Module — nRF24L01**Role:**

Wireless communication between remote and main unit

Why selected:

- Address-based communication → enables pairing
- Low power consumption
- Widely available and cost-effective
- SPI interface compatible with MCU

3.7 PIR Sensor Front-End — NCS36000 + PIR Element**Role:**

Motion detection

Why selected:

- Integrated amplification and filtering
- Reduces need for discrete analog circuitry
- Provides clean digital output to MCU
- Includes enable pin for remote-based sensor control

3.8 Ambient Light Sensor — LDR (GL5528)

Role:

Detects ambient light level

Why selected:

- Low cost and simple implementation
- Wide resistance range → easy threshold detection
- Suitable for day/night discrimination

3.9 LED Selection — Samsung LM281B+

Role:

Illumination source

Why selected:

- High efficiency (>120 lm/W)
- Designed for lighting (not indicator use)
- Meets 20–30 lumen target within 0.25W budget
- Wide viewing angle for uniform light distribution

3.10 LED Drive Strategy — Resistor + MOSFET

Role:

Controls LED current and switching

Why selected over constant-current / buck ICs:

- Works across full battery voltage range (4.2V → 3.4V)
- No headroom requirement
- No switching noise (important for RF + PIR stability)
- Minimal component count and footprint

Note:

Dedicated constant current ICs (e.g. NSI45020, AL5809) were evaluated but rejected due to insufficient voltage headroom at low battery levels.

4. System Interaction (MAIN vs REMOTE)

Remote PCB

- MCU: ATtiny1616
- RF: nRF24L01
- Inputs: 2 buttons

Function:

- Sends command packets on button press
- Enters low-power sleep otherwise

Power Supply Options:

- CR2032 coin cell OR
- 2× AAA batteries

Justification:

- Direct 3V operation (no regulator needed)
- Simple and compact design

Main PCB

- Receives RF commands
- Updates system state
- Controls LED output accordingly

5. RF Communication & Pairing Logic

Pairing Process

1. **Enter Pairing Mode**
 - User presses pair button on main PCB
 - Device enters pairing window
2. **Remote Initiates Pairing**
 - User presses both buttons simultaneously
 - Remote transmits its unique ID
3. **Address Storage**

- Main PCB receives ID
- Stores in non-volatile memory

4. **Locking**

- System exits pairing mode
- Only responds to stored address

Commands

- LIGHT ON
- LIGHT OFF
- SENSOR ENABLE
- SENSOR DISABLE

6. Control Logic (State Machine)

States

1. **AUTO MODE**
 - Controlled by PIR + LDR
2. **MANUAL ON**
 - Light forced ON (ignores sensors)
3. **SENSOR DISABLED**
 - Sensors ignored
 - Only remote can control light

7. Key Design Advantages

- Safe isolation from mains
- Seamless battery backup
- Stable 3.3V rail for RF and MCU
- Efficient lighting within power budget
- Reliable RF pairing (no interference between devices)
- Compact design suitable for circular PCB

